

Posters

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Spatial Decoding of Lateralized Pain Anticipation from EEG Using Deep Temporal Neural Networks

OBJECTIVES

We present a deep learning framework to decode spatially specific pain anticipation from electroencephalography (EEG) signals recorded during a Pavlovian threat conditioning paradigm. Beyond binary classification of threat versus safety, our primary objective is to determine whether anticipatory neural activity contains sufficient information to distinguish the predicted spatial location of pain (left vs. right arm) prior to stimulus delivery. This approach aims to characterize lateralized anticipatory brain dynamics and evaluate whether data-driven temporal models can capture structured, spatially organized predictive states in cortical activity.

MATERIALS

Thirty healthy adult participants underwent a threat conditioning paradigm comprising acquisition and reversal phases. During acquisition, two visual cues predicted a painful shock to either the left (CS+L) or right (CS+R) arm, while a third cue (CS-) signaled safety. In the reversal phase, contingencies were swapped, requiring flexible updating of spatial predictions. This design enabled assessment of both stability and adaptability of anticipatory neural representations. Peripheral physiological measures (SCR, MEPS) were recorded but analyzed separately.

METHOD

EEG was recorded from 64 scalp electrodes at 5 kHz. Data were artifact-corrected, filtered (0.5–50 Hz), downsampled (500 Hz), baseline-corrected, and z-scored per participant. Epochs included a pre-cue baseline (–1–0 s) and cue interval (0–4.5 s). We trained a Temporal Convolutional Network (TCN) with stacked 1D convolutional layers and global pooling for multi-class classification (CS+L, CS+R, CS-). In parallel, time–frequency representations (STFT spectrograms) were evaluated as model inputs. Training followed a participant-level split (70/15/15%) to ensure generalization across individuals.

RESULTS

Model optimization is ongoing. Preliminary analyses indicate above-chance classification of lateralized pain anticipation during the cue period, suggesting that EEG signals encode spatially organized predictive states prior to nociceptive input. Comparative analyses across raw and time–frequency inputs are in progress.

DISCUSSION

These findings support the hypothesis that anticipatory brain states exhibit structured, lateralized dynamics detectable through deep temporal modeling. The paradigm further allows investigation of neural flexibility during contingency reversal, linking predictive coding and adaptive learning mechanisms.

CONCLUSIONS

This framework demonstrates the feasibility of decoding spatial pain anticipation from EEG and provides a foundation for integrating explainable AI approaches to identify neural features underlying pre

Jobst Heitzig (Potsdam Institute for Climate Impact Research, Germany)

AI systems that empower humans in a complex world

We present results from experiments where AI systems are tasked to empower human agents in multi-agent environments with complex dynamics.

Ali Rana Atilgan (Sabanci University, Istanbul, Turkey)

Trajectory Ensembles to Elucidate Mutant-Induced Kinetic Modulation of Proteins

Many experiments and coarse-grained models of protein conformational change report only a single rate, while the underlying dynamics are high-dimensional, contact-structured, and

heterogeneous. We develop a two-layer sequence–structure framework that uses Maximum Caliber (MaxCal) to lift such coarse kinetic information into an explicit trajectory ensemble on a structurally defined path, and to quantify how mutations reshape that ensemble.

In the first layer, a compact sequence–structure theory estimates mutation-induced changes in gating rates. A Potts model inferred from a multiple-sequence alignment (MSA) is blended with a Gaussian Network Model built on the closed conformation; Potts couplings modulate effective spring constants, deforming the Kirchhoff matrix in a sequence-dependent manner. A gating contribution is extracted from the lowest-frequency mode that best aligns with the closed $\rightarrow$ occluded displacement, and an Arrhenius-like barrier factor accounts for non-harmonic reorganization. Applied to E. coli DHFR, this layer predicts mutant-to-wild-type rate ratios in close agreement with experiment and yields a target closed $\leftrightarrow$ occluded rate for each sequence.

In the second layer, we use two endpoint structures and the target rate to construct a one-dimensional structural path, a contact-breaking barrier profile, and an effective free-energy landscape. MaxCal then infers nearest-neighbor transition rates that obey local detailed balance and reproduce the prescribed mean first-passage time (FPT), yielding the maximum-entropy Markov dynamics consistent with the structural and kinetic constraints. From the resulting ensemble we obtain dwell-time maps, state-resolved fluxes, path-length and FPT distributions, and path entropies that quantify route diversity, kinetic bottlenecks, and gating robustness. Whenever two structures, an MSA, and a few kinetic observables are known, the framework furnishes a minimally biased non-equilibrium process on a constrained graph.

Wolfgang Renz (University of Applied Sciences, Hamburg, Germany)

Combining Machine Learning and Dynamic Network Models for Sepsis Prediction

We enhance short-term sepsis predictions by integrating machine learning techniques like Auto-Encoders and Gated Recurrent Units with a dynamical 2-layer network model of adaptive phase oscillators [1,2] representing the interaction between parenchymal cells (functional organ cells) and the immune system via cytokines. The model trajectories determined by machine learning are used for detection and prediction of critical infection states and mortality. The model-based predictions are compared with those of purely data-based approaches in terms of predictive power and interpretability. To this end we project real high-dimensional medical patient data into the low-dimensional parameter space of the model.

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Georg Reich (TU Berlin, Germany)

CA3 Network Model Learns to Replay Hundreds of Chaotic Attractor Dynamics via Global Remapping and Synaptic Delays

Place cells in the hippocampus are thought to provide a sparse code of locations and abstract variables, and during rest or sleep replay their previous activation sequences for memory consolidation and planning. Animals can encounter many different environments in their life, and learn to predict complex spatiotemporal patterns in different contexts. Global remapping refers to the observation that place field representations are decorrelated between different contextual environments. It has been previously found that delays can provide a powerful embedding of chaotic dynamical systems and enable an "unprecedented memory capacity" in neural circuits. In this work we make use of such transmission delays in a single network of place cells. For a given chaotic system and delay, synaptic weights are learned through the Ulam-Galerkin method where the basis is given by disjoint place fields on the attractor. This learning rule is local and online, and

thus biologically plausible. In our contribution, we extend this approach to multiple delays and multiple learned systems. As a result, our model can learn the attractor dynamics of hundreds of systems and selectively replay each system in response to an initial stimulus. We empirically measure the memory capacity of our model on permutations of place fields on the Lorenz attractor, and make first steps towards an analytical treatment.

Rossella Rizzo (University of Palermo, Italy)

Stationary and oscillatory patterns in a FitzHugh-Nagumo model with anomalous diffusion

Anomalous diffusion phenomena frequently occur in natural systems. Interesting examples include autocatalytic chemical reactions on porous media, the preferential movement of species driven by safety or hunting strategies, and long-range interactions in ion channels within the plasma membrane. Cross-diffusion is a kind of nonlinear diffusion used to describe population dynamics where the gradient of one species induces the flux of the other species [1, 2]. On the other hand, super-diffusive processes, such as Lévy flights, can describe the mass diffusion in plasmas or foraging dynamics of birds and oceanic predators for randomly located resources and lead to fractional derivative modeling [3, 4].

In this talk, we investigate how anomalous diffusion influences the formation of stationary patterns in the FitzHugh-Nagumo model, which represents the paradigm system to describe excitable dynamics both in chemical reactions and population dynamics [5, 6]. We find that introducing anomalous diffusion terms allows for pattern formation in both short-range activation/long-range inhibition or long-range activation/short-range inhibition, relaxing the typical requirement for a rapidly diffusing inhibitor in the case of classical diffusion [7]. Moreover, in the presence of cross-diffusion, the spatial structures induced by long-range activation/short-range inhibition mechanisms are always out of phase (cross-Turing patterns) and subcritical in most of the instability regions [8].

Finally, using the formalism of amplitude equations, we derive the asymptotic profiles of the stationary solutions and classify the bifurcation, distinguishing between super- and subcritical transitions. Moreover, we investigate the dynamics near a co-dimension 2 Turing-Hopf bifurcation point, and find the conditions under which we can have oscillatory Turing patterns.

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Merten Stender (TU Berlin, Germany)

Evolution strategies for optimal minimal reservoir computers

Ronja Strömsdörfer (TU Berlin, Germany)

Spike-frequency and h-current based adaptation as candidate modulators of the propagation of traveling waves

During non-REM sleep, the brain shows repetitive patterns of slow oscillations (SOs, <1 Hz) between periods of active (up-state) and silent (down-state) neuronal activity, which are assumed to play a crucial role in memory consolidation (Rasch and Born, 2013).

Cortical whole-brain modeling studies found the connectome to be a key driver of propagation features (e.g., directionality) of these waves (Koller et al., 2024; Cakan et al., 2022) while intrinsic mechanisms are also candidate modulators of propagation features (see Alegre-Cortés et al. (2025), who showed in-silico that changes in adaptation affected neighboring neuronal populations). Both spike-frequency adaptation (SFA) and hyperpolarization-activated (h-)current based adaptation were experimentally and computationally shown to drive traveling waves of SO-like dynamics (Dalla Porta et al., 2024; Mehrotra et al., 2024) while their modulatory contribution to propagation features remains unclear. Mechanistically contrary, meaning reduced to their canonical roles as negative feedback activated in high-activity states (SFA) vs. positive feedback activated in low-activity states (h-currents), they are dynamically equivalent under transformation and compensation (Strömsdörfer and Obermayer, 2025).

Building on the publication of Strömsdörfer and Obermayer (2025), we implement an adaptive Wilson–Cowan field model incorporating both mechanisms with equivalency-breaking properties on a two-dimensional spatial domain. We investigate how these equivalency-breaking properties differentially affect propagation features of traveling waves, and how their effects compare to recurrence weights. Preliminary results indicate that spatial gradients in adaptation strength induce directionality independent of initialization, and that gradient steepness increases wave speed.

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Simon Vock (Charité Berlin, Germany)

Criticality governs deep learning: enhancing performance, enabling continual learning, mitigating model collapse

The rapid advances in artificial intelligence (AI) have largely been driven by scaling deep neural networks (DNNs) - increasing model size, data, and computational resources. However, performance is ultimately governed by network dynamics. The lack of a principled understanding of DNN dynamics beyond heuristic-based design has contributed to challenges with their robustness, suboptimal performance, high energy consumption and pathologies in continual and AI-generated content learning. Increasing evidence suggests that the human brain may avoid these problems by operating at a critical phase transition. Inspired by this principle, we here propose that criticality may provide a unifying framework linking structure, dynamics, and function also in DNNs. First, by analyzing more than 80 state-of-the-art DNN models and well established benchmark datasets (ImageNet, MNIST), we report that a decade of AI progress has implicitly driven successful networks towards criticality, as measured by largest Lyapunov exponents λ_0 (Spearman $r = -0.5$, $p < 10^{-4}$), explaining why certain architectures succeeded while others failed. Second, incorporating criticality explicitly into training improves even highly optimized models by more than 0.4 percentage points ($p < 0.001$) and establishes robustness across weight initializations, preventing key limitations of current models. Third, we show that catastrophic AI pathologies, including continual learning degradation and model collapse from AI-generated training data, constitute a loss of critical dynamics. By keeping networks critical, we provide the first principled solution to these fundamental AI problems by mitigating performance degradation and model collapse. This work confirms criticality as substrate-independent principle of intelligence, connecting AI advancement with core principles of brain function. For systems neuroscience, this work offers a link between optimal function in biological and artificial neural networks. For AI, it provides profound theoretical insights along with immediate practical value solving major challenges to ensure long-term DNN performance and resilience as models grow in scale and complexity.

Yu Wang (Potsdam Institute for Climate Impact Research, Germany)

Universal dynamic phenomena in delay systems: fundamentals and applications

The majority of delay differential equations (DDEs) with one delay have only a few bifurcation scenarios, which can be explicitly described. We explore absolute stability and universal bifurcation scenarios in DDEs using asymptotic continuous spectrum (ACS) theory. We then combine the Master Stability Function (MSF) method for discussing the dynamics of active-agent systems. First, we present how universality classes and Hopf bifurcation sequences in single-delay DDEs can be characterized through the ACS, reveal general transversality results in the large-delay regime, and consider the three most common universality classes. For each of them, we explicitly describe the sequence of stabilizing and destabilizing bifurcations. Then, we apply the above framework to active-agent systems with inertia and delayed feedback, analyzing stability conditions for formation patterns in both uncoupled and coupled settings. These results provide a unified perspective on stable coordination, pattern formation, and universal delay-induced dynamics in complex systems. Additionally, we investigate interactions in the large-delay limit, where delays affect inter-agent coupling, while local feedback remains instantaneous. In this limit, we prove rigorously that the stability region in the complex plane of the eigenvalues of the Laplacian matrix converges to a circle centered at the origin, a phenomenon previously observed in delay-coupled networks. Our findings provide a universal framework for understanding stable formations and motions of active agents with delayed interactions.

Dan Zhao (Potsdam Institute for Climate Impact Research, Potsdam, Germany)

Spatial heterogeneity of frequency stability in the Italian power grid under localized intermittent perturbations

We study how localized intermittent perturbations affect frequency stability in a 127-node model of the Italian high-voltage transmission grid. Using the swing equation with bounded wind-like forcing applied to individual generators, we construct generator-wise vulnerability maps from the ensemble-averaged time-windowed frequency dispersion. The response is strongly heterogeneous across generators in both strongly coupled and near-critical regimes. Although the absolute response level varies with forcing conditions, the spatial ordering of the most vulnerable generators remains comparatively robust across changes in the intermittency parameter, indicating that perturbation location is a dominant determinant of vulnerability. Spatiotemporal analysis further reveals selective disturbance spreading through specific network regions rather than spatially homogeneous amplification. Near critical coupling, the interpretation of vulnerability becomes strongly protocol dependent. An abrupt quench of the coupling strength can already produce persistent desynchronization in the absence of stochastic forcing, whereas an adiabatic continuation protocol provides a cleaner reference state for isolating the additional effect of intermittent perturbations. These findings provide a basis for identifying critical injection sites and informing spatially targeted strategies for maintaining frequency stability in power grids with increasing renewable penetration.